LEVEL UP ACADEMY

Class 10 | CBSE | Science

Pre-Board Mock 1 (CBSE pattern)

Chapter / Topic: Full syllabus (Chapters 1–13)

Name: _____ Roll No.: _____ Date: _____

Time: 3 hours **Max. Marks: 80**

General instructions:

1. This paper has 39 questions in 5 sections (A–E). All questions are compulsory. Internal choice is given in some questions.
2. Section A: Q1–Q20, 1 mark each (MCQs and Assertion–Reason). Section B: Q21–Q26, 2 marks each. Section C: Q27–Q33, 3 marks each.
3. Section D: Q34–Q36, 5 marks each. Section E: Q37–Q39, case-based, 4 marks each.
4. Write balanced equations, show formulae and units in numericals, and draw neat labelled diagrams wherever required.
5. This mock follows the reported 2026-27 pattern. Check the official CBSE sample paper for any changes.

SECTION A

20 marks

Q1–Q16: choose the correct option. Q17–Q20: Assertion–Reason. For these, choose (a) both A and R true and R explains A; (b) both true but R does not explain A; (c) A true, R false; (d) A false, R true.

Q1. Which of the following is a combination reaction? **[1]**

(a) $CaO + H_2O \rightarrow Ca(OH)_2$ (b) $2KClO_3 \rightarrow 2KCl + 3O_2$
(c) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ (d) $NaOH + HCl \rightarrow NaCl + H_2O$

Q2. The pH of a solution is 13. The solution is: **[1]**

(a) strongly acidic (b) weakly acidic
(c) strongly basic (d) neutral

Q3. Which of these metals does NOT give hydrogen gas with dilute hydrochloric acid? **[1]**

(a) Copper (b) Magnesium
(c) Zinc (d) Iron

Q4. The name of the compound CH_3CHO is: **[1]**

(a) methanal (b) ethanol
(c) ethanal (d) ethanoic acid

Q5. Which of these has ten molecules of water of crystallisation? **[1]**

(a) Gypsum (b) Copper sulphate crystals
(c) Plaster of Paris (d) Washing soda

Q6. A few drops of phenolphthalein are added to sodium hydroxide solution. The colour of the solution becomes: **[1]**

(a) red (b) pink
(c) colourless (d) yellow

Q7. Bile is poured into which part of the alimentary canal? **[1]**

(a) Stomach (b) Small intestine
(c) Large intestine (d) Oesophagus

Q8. The correct path of urine in the human body is: **[1]**

(a) Kidney → urethra → urinary bladder → ureter
(b) Kidney → urinary bladder → ureter → urethra
(c) Kidney → ureter → urinary bladder → urethra
(d) Urinary bladder → kidney → ureter → urethra

Q9. Which hormone prepares the body to deal with an emergency? **[1]**

(a) Adrenaline (b) Insulin
(c) Thyroxin (d) Growth hormone

Q10. Vegetative propagation by leaves is seen in: **[1]**

(a) potato (b) rose
(c) sugarcane (d) Bryophyllum

Q11. A cross between a heterozygous tall pea plant (Tt) and a short plant (tt) gives tall plants in the proportion: **[1]**

(a) 25% (b) 50%
(c) 75% (d) 100%

Q12. Which of the following is non-biodegradable? **[1]**

(a) Cow dung (b) Paper
(c) Fruit peels (d) DDT

Q13. A concave mirror forms a real image of the same size as the object. The object is placed: **[1]**

(a) at F (b) at C
(c) between F and C (d) beyond C

Q14. A wire of resistance 10 Ω is stretched uniformly to double its length. Its new resistance is: **[1]**

(a) 5 Ω (b) 10 Ω
(c) 20 Ω (d) 40 Ω

Q15. Which colour of white light is deviated the most by a glass prism? **[1]**

(a) Red (b) Yellow
(c) Violet (d) Green

Q16. The magnetic field lines inside a long, current-carrying solenoid are: **[1]**

(a) parallel straight lines (b) concentric circles
(c) curved outwards (d) absent

Q17. **Assertion (A):** Rancidity of food can be slowed down by keeping it in airtight containers.
Reason (R): Airtight containers reduce contact with oxygen, slowing the oxidation of fats and oils. **[1]**

Q18. **Assertion (A):** Pepsin acts on proteins in the stomach.
Reason (R): The gastric glands secrete hydrochloric acid, which makes the medium in the stomach acidic, as needed for pepsin to act. **[1]**

Q19. **Assertion (A):** A convex lens can form both real and virtual images.
Reason (R): A convex lens always forms the image on the side opposite to the object. **[1]**

Q20. **Assertion (A):** The concentration of pesticides is highest in organisms at the top of a food chain.
Reason (R): Pesticides are non-biodegradable and build up at each trophic level. **[1]**

SECTION B

12 marks

Q21. What happens when silver chloride is kept in sunlight? Write the balanced equation and name the type of reaction. **[2]**

Q22. In the electrolysis of water, name the gases collected at the cathode and the anode. Why is the volume of one gas double that of the other? **[2]**

Q23. Give the function of: (a) villi in the small intestine, (b) salivary amylase. **[2]**

Q24. Why do gametes have only half the number of chromosomes found in body cells? **[2]**

OR

Why does menstruation occur?

Q25. A 6 Ω and a 3 Ω resistor are connected in parallel, and this combination is connected in series with a 4 Ω resistor and a 6 V battery. Find the total resistance and the current drawn from the battery. **[2]**

OR

The far point of a myopic person is 80 cm in front of the eye. What is the nature and power of the lens needed to correct the problem?

Q26. Why are some substances biodegradable and others non-biodegradable? Give one example of each. **[2]**

SECTION C

21 marks

Q27. Describe the electrolytic refining of copper. **[3]**

Q28. A compass needle is placed near a straight wire carrying current. **[3]**

(i) Why is the needle deflected? **[1]**

(ii) What happens to the deflection if the current is increased? **[1]**

(iii) What happens to the deflection if the compass is moved farther from the wire? **[1]**

Q29. An organic compound A with molecular formula $C_2H_4O_2$ turns blue litmus red. It reacts with sodium carbonate to give a gas that turns lime water milky. Identify A and write the balanced equations for both reactions. **[3]**

Q30. Describe the path of blood through the human heart, showing how double circulation works. **[3]**

Q31. You accidentally touch a hot plate and pull your hand away at once. Describe the reflex arc involved. **[3]**

OR

Give three differences between nervous control and hormonal control.

Q32. A 2 cm tall object is placed 30 cm from a convex lens of focal length 20 cm. Find the position, size and nature of the image. **[3]**

OR

A concave mirror produces a real image three times the size of an object placed 10 cm in front of it. Where is the image? Find the focal length of the mirror.

Q33. An electric heater is rated 1000 W, 220 V. **[3]**

(i) Find its resistance. **[1]**

(ii) Find the current it draws. **[1]**

(iii) How many units (kWh) does it use if run for 2 hours a day for 30 days? **[1]**

SECTION D

15 marks

Q34. Answer the following: **[5]**

(i) Differentiate between saturated and unsaturated hydrocarbons, with one example each. How can you tell them apart by burning? **[3]**

(ii) Write the equation for the addition of hydrogen to ethene and state the condition needed. **[2]**

OR

(a) Describe an activity to show that acids produce H^+ ions only in water, using solutions of HCl, glucose and alcohol. [3]
(b) Why does distilled water not conduct electricity, while rain water does? [2]

Q35. Answer the following: **[5]**

(i) Name the parts of the human male reproductive system and give the function of each. **[3]**

(ii) Give two changes seen in boys at puberty. **[2]**

OR

(a) Explain how Amoeba obtains and digests its food. [3]
(b) Name the raw materials for photosynthesis, and say where a plant gets each of them. [2]

Q36. Answer the following: **[5]**

(i) Derive an expression for the equivalent resistance of three resistors R_1, R_2 and R_3 connected in series. **[3]**

(ii) A 4 Ω and a 12 Ω resistor are connected in parallel, and the combination is in series with a 3 Ω resistor across a 12 V battery. Find the total current. **[2]**

OR

(a) Derive an expression for the equivalent resistance of three resistors connected in parallel. [3]
(b) Why is tungsten used to make the filaments of electric bulbs? [2]

SECTION E

12 marks

Read each passage and answer the questions that follow.

Q37. Baking soda (sodium hydrogencarbonate) is commonly used in kitchens. When heated, it decomposes and releases carbon dioxide. If only baking soda is used in a cake, the cake rises but tastes bitter. So baking powder, a mixture of baking soda and a mild edible acid such as tartaric acid, is used instead. **[4]**

(i) Write the chemical formula of baking soda. **[1]**

(ii) Write the balanced equation for its decomposition on heating. **[1]**

(iii) Why does a cake made with only baking soda taste bitter, and how does tartaric acid help? **[2]**

Q38. A student covered a healthy potted plant with a transparent polythene bag and tied it around the stem. After a few hours in sunlight, water droplets appeared on the inside of the bag. **[4]**

(i) Name the process that caused the droplets. **[1]**

(ii) Through which structures does most of this water leave the plant? **[1]**

(iii) Give two ways in which this process helps the plant. **[2]**

Q39. Spherical mirrors are used widely in daily life. Dentists and people shaving use one type of mirror. Vehicles use another type as rear-view mirrors because it gives a wider view of the traffic behind. **[4]**

(i) Which mirror is used as a shaving mirror, and why? **[1]**

(ii) Why is a convex mirror preferred as a rear-view mirror? **[1]**

(iii) A convex rear-view mirror has a radius of curvature of 3.00 m. A bus is 5.00 m behind it. Find the position of the image. **[2]**

~ All the best for your boards! ~